Sentinel Post-Mortem: Database Host Resource Pressure

Incident type: Simulated infrastructure health event | Evidence: supplied system_telemetry.db | Status: anomaly detected; root cause not established

1. Executive summary

Sentinel identified resource pressure on SRV-002 (db-prod-1, US-West): CPU 88.5% and memory 92.3%, exceeding default thresholds of 85% and 90%. The remediation safety gate blocked high-risk actions because CPU load exceeded the 85% critical threshold. No destructive command was run.

2. Data quality and scope

The brief anticipates servers, logs, and incidents tables. The delivered database contains only servers (2 rows) and logs (5,000 rows); incidents is absent. All log messages are the generic 'Sample log message', all belong to SRV-001, and all identify Auth-Service. The records cannot explain SRV-002's resource pressure. Log levels: ERROR 1,614; WARN 1,686; INFO 1,700.

3. Detection

Server	CPU	Memory	Status	Region
web-prod-1 / SRV-001	45.2%	60.1%	ACTIVE	US-East
db-prod-1 / SRV-002	88.5%	92.3%	ACTIVE	US-West

4. SQL investigation

```
SELECT * FROM servers ORDER BY server_id;\n\nSELECT log_level, COUNT(*) AS count\nFROM logs GROUP BY log_
```

5. Diagnosis

Combined CPU and memory pressure may reflect workload saturation, expensive queries, connection accumulation, cache growth, or a recent deployment. These remain hypotheses. Query-level metrics, connection counts, process metrics, and deployment history are not present, so a definitive root cause cannot be assigned.

6. Remediation and verification

At 88.5% recorded CPU, the remediation agent blocks system_reboot, shutdown, format_disk, and kill_database. It recommends operator-led inspection of top queries, connection-pool utilization, memory/cache trends, and recent deployment changes. Verification re-queries telemetry and confirms the threshold breach remains. Because this implementation is read-only/simulated, recovery is not claimed; failed verification routes back to diagnosis and escalates after the retry limit.

7. Outcome

Detected: high CPU and memory on db-prod-1. **Contained:** high-risk actions blocked. **Recovered:** not demonstrated. **Root cause:** undetermined from available evidence.

8. NL2SQL trace example

Question: Show server health

```
SELECT server_id, hostname, region, status,\n       cpu_usage_percent, memory_usage_percent\nFROM servers
```

An out-of-schema question such as 'What is the CEO's favorite color?' receives a graceful unsupported-schema response and executes no SQL. LangSmith trace capture instructions are provided in TRACE_CAPTURE.md. No hosted trace screenshot is fabricated because credentials and an actual LangSmith run were not supplied.